

MH2100 Calculus III

Midterm II (Solutions)

AY 2018/2019, Semester 1

Quantitative Research Society @NTU

November 2018

Overview

- **Historical paper:** the original paper was held in November 2018 with **90 minutes** allowed.
- **Original instructions:** answer all three questions; candidates could write anywhere on the paper; one two-sided A4 page of handwritten notes was permitted.
- **2026 scope note:** the current midterm scope ends at double integrals, including polar coordinates. General Jacobian substitutions, triple integrals, cylindrical and spherical coordinates, and line integrals are beyond that scope. They are retained and solved here because they appear in the historical paper.

Instructions to candidates (historical)

1. DO NOT TURN OVER PAPER UNTIL INSTRUCTED.
2. This midterm paper contains THREE (3) questions.
3. Answer ALL questions. The marks for each question are indicated at the beginning of each question.
4. Candidates can write anywhere on this midterm paper.
5. A two-sided A4 piece of paper with handwritten notes is permitted.

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Question 1 (40 Marks)

(beyond 2026 midterm scope): Parts (iv)–(vii) use triple integrals, cylindrical/spherical coordinates, or a general Jacobian substitution.

Circle (only) the correct answer for each question. Each question is worth 5 marks.

- (i) Calculate the double integral $\iint_R (6x^2y^3 - 10x^4) \, dA$ where $R = \{(x, y) \mid 0 \leq x \leq 1, 0 \leq y \leq 4\}$.
- (a) 190
(b) 170
(c) 130
(d) 150
(e) 120
- (ii) Use polar coordinates to find the volume of the solid under the paraboloid $z = x^2 + y^2$ and above the disk $x^2 + y^2 \leq 9$ (in the plane $z = 0$).
- (a) 81π
(b) 162π
(c) 324π
(d) 27π
(e) None of the above
- (iii) Evaluate the iterated integral by converting to polar coordinates:

$$\int_{-5}^5 \int_0^{\sqrt{25-y^2}} (x^2 + y^2)^{3/2} \, dx \, dy.$$

- (a) 625π
(b) 1250π
(c) $\frac{625\pi}{2}$
(d) $\frac{625\pi}{4}$
(e) None of the above
- (iv) Evaluate the iterated integral:

$$\int_0^\pi \int_0^1 \int_0^{\sqrt{1-y^2}} 4y \sin x \, dz \, dy \, dx.$$

- (a) $\frac{8}{3}$
(b) 4
(c) $\frac{4}{3}$
(d) $\frac{2}{3}$

- (e) None of the above
- (v) Use cylindrical coordinates to evaluate $\iiint_T \sqrt{x^2 + y^2} \, dV$, where T is the solid bounded by the cylinder $x^2 + y^2 = 1$ and the planes $z = 4$ and $z = 8$.
- (a) $\frac{8\pi}{3}$
- (b) 4π
- (c) $\frac{4\pi}{3}$
- (d) $\frac{2\pi}{3}$
- (e) None of the above
- (vi) Use spherical coordinates to evaluate $\iiint_B (x^2 + y^2 + z^2)^2 \, dV$ where B is the ball with center the origin and radius 4.
- (a) $\frac{65536\pi}{7}$
- (b) $\frac{4096\pi}{5}$
- (c) $\frac{8192\pi}{3}$
- (d) $\frac{32768\pi}{7}$
- (e) None of the above
- (vii) Find the Jacobian of the plane transformation given by $x = 5u \sin v$ and $y = 4u \cos v$.
- (a) $\frac{\partial(x, y)}{\partial(u, v)} = 9u$
- (b) $\frac{\partial(x, y)}{\partial(u, v)} = 36u$
- (c) $\frac{\partial(x, y)}{\partial(u, v)} = -20u$
- (d) $\frac{\partial(x, y)}{\partial(u, v)} = -u$
- (e) None of the above
- (viii) Evaluate the iterated integral:
- $$\int_1^3 \int_y^3 12xy \, dx \, dy.$$
- (a) 96
- (b) 116
- (c) 106
- (d) 126
- (e) None of the above

Solution**Method 1: Direct computations**

(i)

$$\begin{aligned}\int_0^1 \int_0^4 (6x^2y^3 - 10x^4) dy dx &= \int_0^1 (384x^2 - 40x^4) dx \\ &= 128 - 8 = 120 \implies \boxed{\text{Answer (e): } 120.}\end{aligned}$$

(ii) Polar: $0 \leq r \leq 3$, $0 \leq \theta \leq 2\pi$, $z = r^2$, $dA = r dr d\theta$.

$$V = \int_0^{2\pi} \int_0^3 r^3 dr d\theta = 2\pi \cdot \frac{81}{4} = \frac{81\pi}{2} \implies \boxed{\text{Answer (e): None of the above.}}$$

(iii) Right half-disk: $-\pi/2 \leq \theta \leq \pi/2$, $0 \leq r \leq 5$, $(x^2 + y^2)^{3/2} = r^3$.

$$\int_{-\pi/2}^{\pi/2} \int_0^5 r^3 \cdot r dr d\theta = \pi \left[\frac{r^5}{5} \right]_0^5 = 625\pi \implies \boxed{\text{Answer (a): } 625\pi.}$$

(iv) $\int_0^{\sqrt{1-y^2}} dz = \sqrt{1-y^2}$.

$$4 \int_0^\pi \sin x dx \int_0^1 y \sqrt{1-y^2} dy = 4(2) \left[-\frac{(1-y^2)^{3/2}}{3} \right]_0^1 = \frac{8}{3} \implies \boxed{\text{Answer (a): } \frac{8}{3}.}$$

(v) Cylindrical: $\sqrt{x^2 + y^2} = r$, $0 \leq r \leq 1$, $0 \leq \theta \leq 2\pi$, $4 \leq z \leq 8$, $dV = r dr d\theta dz$.

$$\int_4^8 \int_0^{2\pi} \int_0^1 r \cdot r dr d\theta dz = 4(2\pi) \frac{1}{3} = \frac{8\pi}{3} \implies \boxed{\text{Answer (a): } \frac{8\pi}{3}.}$$

(vi) Spherical: $(x^2 + y^2 + z^2)^2 = r^4$, $dV = r^2 \sin \phi dr d\phi d\theta$.

$$\begin{aligned}\int_0^{2\pi} \int_0^\pi \int_0^4 r^6 \sin \phi dr d\phi d\theta &= 2\pi \cdot 2 \cdot \frac{4^7}{7} \\ &= \frac{65536\pi}{7} \implies \boxed{\text{Answer (a): } \frac{65536\pi}{7}.}\end{aligned}$$

(vii)

$$\begin{aligned}\frac{\partial(x, y)}{\partial(u, v)} &= \begin{vmatrix} 5 \sin v & 5u \cos v \\ 4 \cos v & -4u \sin v \end{vmatrix} \\ &= -20u(\sin^2 v + \cos^2 v) = -20u \implies \boxed{\text{Answer (c): } -20u.}\end{aligned}$$

(viii)

$$\int_1^3 \int_y^3 12xy dx dy = \int_1^3 6y(9 - y^2) dy = \left[27y^2 - \frac{3}{2}y^4 \right]_1^3 = 96 \implies \boxed{\text{Answer (a): } 96.}$$

Method 2: symmetry (part (viii))

$T = \{1 \leq y \leq x \leq 3\}$ is half of $[1, 3]^2$ cut by $x = y$, and $12xy$ is symmetric in $x \leftrightarrow y$:

$$\iint_T 12xy \, dA = \frac{1}{2} \iint_{[1,3]^2} 12xy \, dA = 6 \left(\int_1^3 x \, dx \right)^2 = 6 \cdot 4^2 = 96.$$

Unofficial QRS marking scheme (40 marks)

- (i) **(5 marks)** 2 marks for correct integration in y ; 2 for integration in x ; 1 for 120 and answer (e).
- (ii) **(5 marks)** 2 marks for polar volume setup; 2 for correct evaluation; 1 for recognising that $81\pi/2$ is answer (e).
- (iii) **(5 marks)** 2 marks for the half-disk bounds; 2 for the polar integrand including the Jacobian; 1 for answer (a).
- (iv) **(5 marks)** 2 marks for integrating in z ; 2 for the remaining integrations; 1 for answer (a).
- (v) **(5 marks)** 2 marks for cylindrical bounds; 2 for the integrand and volume factor; 1 for answer (a).
- (vi) **(5 marks)** 2 marks for spherical conversion; 2 for all bounds and integrations; 1 for answer (a).
- (vii) **(5 marks)** 3 marks for the determinant; 1 for simplification; 1 for answer (c).
- (viii) **(5 marks)** 3 marks for the inner integral; 1 for the outer integral; 1 for answer (a).

Question 2**(30 Marks)**

(beyond 2026 midterm scope): Part (b) uses a general Jacobian substitution and part (c) is a triple integral.

- (a) Evaluate the iterated integral

$$\int_0^4 \int_{\sqrt{y}}^2 e^{x^3} dx dy.$$

- (b) Use the transformation $x = \sqrt{2}u - \sqrt{\frac{2}{3}}v$; $y = \sqrt{2}u + \sqrt{\frac{2}{3}}v$ to evaluate the integral

$$\iint_R x^2 - xy + y^2 dA,$$

where R is the region bounded by the ellipse $x^2 - xy + y^2 = 2$.

- (c) Evaluate the triple integral

$$\iiint_R z^2 dV,$$

where R is the solid bounded by the sphere $x^2 + y^2 + z^2 = 1$, the cylinder $x^2 + y^2 = 1$, and the plane $z = 2$.

Solution**Method 1: Reverse the order, then transform**

- (a) Region $\{0 \leq y \leq 4, \sqrt{y} \leq x \leq 2\} = \{0 \leq x \leq 2, 0 \leq y \leq x^2\}$ (Fubini).

$$\begin{aligned} \int_0^2 \int_0^{x^2} e^{x^3} dy dx &= \int_0^2 x^2 e^{x^3} dx \\ &= \frac{1}{3} \int_0^8 e^u du \quad (u = x^3) \implies \boxed{\frac{e^8 - 1}{3}}. \end{aligned}$$

- (b) $a = \sqrt{2}u$, $b = \sqrt{2/3}v$, $x = a - b$, $y = a + b$.

$$\begin{aligned} x^2 - xy + y^2 &= a^2 + 3b^2 = 2u^2 + 2v^2 & (R \mapsto D = \{u^2 + v^2 \leq 1\}) \\ \left| \frac{\partial(x, y)}{\partial(u, v)} \right| &= \left| \begin{vmatrix} \sqrt{2} & -\sqrt{2/3} \\ \sqrt{2} & \sqrt{2/3} \end{vmatrix} \right| = \frac{4}{\sqrt{3}} \\ \iint_R (x^2 - xy + y^2) dA &= \frac{4}{\sqrt{3}} \iint_D 2(u^2 + v^2) du dv \\ &= \frac{8}{\sqrt{3}} \int_0^{2\pi} \int_0^1 \rho^3 d\rho d\theta & (u = \rho \cos \theta, v = \rho \sin \theta) \\ &= \boxed{\frac{4\pi}{\sqrt{3}}} \end{aligned}$$

- (c) The wording is ambiguous: $z = 2$ does not meet the sphere ($z \leq 1$ on it). Take the bounded component above the upper hemisphere:

$$0 \leq r \leq 1, 0 \leq \theta \leq 2\pi, \sqrt{1-r^2} \leq z \leq 2.$$

$$\iiint_R z^2 dV = \int_0^{2\pi} \int_0^1 \int_{\sqrt{1-r^2}}^2 z^2 r dz dr d\theta$$

$$\begin{aligned} &= \frac{2\pi}{3} \int_0^1 r(8 - (1 - r^2)^{3/2}) \, dr \\ &= \frac{2\pi}{3} \left(4 - \frac{1}{5} \right) = \boxed{\frac{38\pi}{15}} \quad \left(\int_0^1 r(1 - r^2)^{3/2} \, dr = \frac{1}{5} \right) \end{aligned}$$

Unofficial QRS marking scheme (30 marks)

- (a) **(10 marks)** 3 marks for the reversed region; 3 for the reversed integral; 3 for the substitution $u = x^3$; 1 for the final value.
- (b) **(10 marks)** 3 marks for diagonalising the ellipse; 3 for the Jacobian; 3 for the unit-disk polar integral; 1 for $4\pi/\sqrt{3}$.
- (c) **(10 marks)** 2 marks for identifying the literal ambiguity; 3 for an explicit bounded-component interpretation and bounds; 3 for the cylindrical integral; 2 for $38\pi/15$ under that interpretation.

Question 3 (30 Marks)

(beyond 2026 midterm scope): All parts involve vector fields or line integrals.

- (a) Find the work done by the force field $\mathbf{F}(x, y) = x \sin(y)\mathbf{i} + y\mathbf{j}$ on a particle that moves along the curve $y = x^2$ from $(0, 0)$ to $(\sqrt{\pi}, \pi)$.
- (b) Evaluate the line integral $6 \int_C yz \cos x \, ds$ where C has parametrisation $x = t$, $y = \cos t$ and $z = \sin t$, with $0 \leq t \leq \pi$.
- (c) Let $\mathbf{F}(x, y) = x^3 y^2 \mathbf{i} + y \left(\frac{x^4}{2} - 2 \right) \mathbf{j}$ and let C be a curve with parametrisation $\mathbf{r}(t) = \sqrt{t} \mathbf{i} + (1 + t^3) \mathbf{j}$ with $0 \leq t \leq 1$ (starting at $\mathbf{r}(0)$ and terminating at $\mathbf{r}(1)$).
- (i) Find a function f such that $\mathbf{F} = \nabla f$.
- (ii) Hence, or otherwise, evaluate the line integral $\int_C \mathbf{F} \cdot d\mathbf{r}$.

Solution

Method 1: Parametrise directly; use a potential where available

- (a) $\mathbf{r}(t) = (t, t^2)$, $0 \leq t \leq \sqrt{\pi}$, $\mathbf{r}' = (1, 2t)$.

$$\mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) = t \sin(t^2) + 2t^3$$

$$W = \int_0^{\sqrt{\pi}} (t \sin(t^2) + 2t^3) \, dt = \left[-\frac{1}{2} \cos(t^2) + \frac{1}{2} t^4 \right]_0^{\sqrt{\pi}} = \boxed{1 + \frac{\pi^2}{2}}$$

- (b) $\mathbf{r}' = (1, -\sin t, \cos t)$, $ds = \|\mathbf{r}'\| \, dt = \sqrt{2} \, dt$.

$$6 \int_C yz \cos x \, ds = 6\sqrt{2} \int_0^{\pi} \cos t \sin t \cos t \, dt = 6\sqrt{2} \left[-\frac{\cos^3 t}{3} \right]_0^{\pi} = 6\sqrt{2} \cdot \frac{2}{3} = \boxed{4\sqrt{2}}$$

- (c) (i)

$$f_x = x^3 y^2 \implies f = \frac{x^4 y^2}{4} + h(y)$$

$$f_y = y \left(\frac{x^4}{2} - 2 \right) \implies h'(y) = -2y \implies \boxed{f(x, y) = \frac{x^4 y^2}{4} - y^2}$$

- (ii) $\mathbf{r}$ is not C^1 at $t = 0$; $t = s^2$ gives the C^1 parametrisation $(s, 1 + s^6)$, $0 \leq s \leq 1$, from $(0, 1)$ to $(1, 2)$. By the fundamental theorem for line integrals ($\mathbf{F} = \nabla f$),

$$\int_C \mathbf{F} \cdot d\mathbf{r} = f(1, 2) - f(0, 1) = (1 - 4) - (-1) = \boxed{-2}$$

Unofficial QRS marking scheme (30 marks)

- (a) **(10 marks)** 3 marks for a valid parametrisation; 3 for $\mathbf{F} \cdot \mathbf{r}'$; 3 for integration; 1 for $1 + \pi^2/2$.
- (b) **(10 marks)** 3 marks for $ds = \sqrt{2} \, dt$; 3 for substitution into the scalar line integral; 3 for integration; 1 for $4\sqrt{2}$.
- (c) **(10 marks)**
- (i) 5 marks: 3 for integrating f_x and 2 for determining $h(y)$.
- (ii) 5 marks: 2 for endpoints, 2 for the potential-difference theorem, and 1 for -2 .